

Expected value, variance, standard deviation

1. Consider the chance experiment of rolling a red die and a green die. Let X be the random variable that assigns to each outcome the higher of the two rolls. For example, $X(5, 3) = 5$, $X(2, 2) = 2$, and $X(1, 6) = 6$. You have calculated the distribution of this random variable. The possible values of X are 1, 2, 3, 4, 5, 6, and X takes on these values with the following probabilities: $P(X = 1) = 1/36$, $P(X = 2) = 3/36$, $P(X = 3) = 5/36$, $P(X = 4) = 7/36$, $P(X = 5) = 9/36$, $P(X = 6) = 11/36$.

Calculate $E[X]$, the expected value of the higher of the two rolls.

Calculate $\text{Var}(X)$, the variance of X .

Calculate $\sigma(X)$, the standard deviation of X .

2. Consider the chance experiment of flipping a coin five times. Let N be the random variable that assigns to each outcome the number of heads. For example, $N(\text{HTHHT}) = 3$. The possible values of N are 0, 1, 2, 3, 4, 5, and N takes on these values with the following probabilities: $P(N = 0) = 1/32$, $P(N = 1) = 5/32$, $P(N = 2) = 10/32$, $P(N = 3) = 10/32$, $P(N = 4) = 5/32$, $P(N = 5) = 1/32$.

Calculate $E[N]$, the expected value of the numbers of H's.

Calculate $\text{Var}(N)$, the variance of N .

Calculate $\sigma(N)$, the standard deviation of N .